Solution Requirements

Date	30 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the Credit Card Approval Prediction system before submitting application details.	High
2	Authorization levels	Applicant can submit and view prediction results. Admin can manage the machine learning model and dataset.	High

3	External interfaces	The system interacts with the trained machine learning model (model.pkl), scaler (scaler.pkl), label encoders (label_encoders.pkl), and dataset (clean_data.csv).	High
4	Transactions processing	The system collects applicant details, preprocesses the data, predicts approval status, and displays the result instantly.	High
5	Reporting	The system displays the prediction result as Credit Card Approved or Credit Card Rejected on the result page.	Medium
6	Business rules, etc.	Applicant information is encoded, scaled, and evaluated using the trained machine learning model before generating the prediction.	High
7	Compliance to laws or regulations	The system protects applicant information and follows data privacy principles by securely handling user data.	Medium
8	*Other*	The system validates user inputs before prediction to reduce invalid or missing data.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction result should be generated quickly after submitting the application.	Response time less than **2 seconds**
2	Scalability	The system should support multiple users without	Supports 100+ concurrent users
3	Security & Data Privacy	Applicant information should be securely processed and not exposed to unauthorized users.	Secure data handling with input validation
4	Reliability & Availability	The system should remain available and provide accurate predictions whenever accessed.	99% uptime
5	Usability & Accessibility	The web application should have a simple, user-friendly interface that works on desktop and mobile browsers.	Easy navigation and responsive design
6	*Other*	The system should allow easy updates to the machine learning model, dataset, and preprocessing files without changing the application structure.	Easy model replacement and maintenance